

MECH5200 Numerical Methods for PDEs : *Video 26:*
A Brief Introduction to Discontinuous Galerkin
Methods (DG-FEM)

David Willis

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1D Advection/Wave Equation

- Consider the 1D advection (1st order wave) equation in strong form:

$$\frac{\partial u}{\partial t} - c \frac{\partial u}{\partial x} = f \quad \text{on } \Omega \quad (1)$$

- With periodic boundary conditions
- Let's start by
- We can define the *weak form*:

$$\left[w \frac{\partial u}{\partial x} \right]_{x_L}^{x_R} - \int_{x_L}^{x_R} \frac{\partial w}{\partial x} \cdot \frac{\partial u}{\partial x} dx = - \int_{x_L}^{x_R} (wf) dx \quad (2)$$

Potential Flow – A Kinetic Energy Minimization Problem

- Let's define the **quadratic form** expression (for flow kinetic energy):

$$J(w) = \frac{1}{2} \int \nabla(w) \cdot \nabla(w) dx - \int f \cdot (w) dx \quad (3)$$

- In the HW problem, $w = \phi$, $\nabla\phi = \vec{velocity}$, and $f = 0$ in the domain. Therefore, this expression loosely describes the flow kinetic energy in the domain Ω :

$$J(w) = \frac{1}{2} \int [\text{velocity}]^T [\text{velocity}] \cdot dx - \int 0 \cdot dx = \frac{K.E.}{m} \quad (4)$$

- We're next going to show solutions ϕ of the PDE minimize the above expression
 - i.e., the solution of the PDE minimizes the kinetic energy of the flow.

$$\phi = \arg \min(J(w)) \quad (5)$$

Potential Flow – A Kinetic Energy Minimization Problem

- Let's introduce a perturbed solution, $w = \phi + v$, in the expression and expand the terms:

$$J(\phi + v) = \frac{1}{2} \int \nabla(\phi + v) \cdot \nabla(\phi + v) dx - \int f \cdot (\phi + v) dx \quad (6)$$

- We can simply expand the terms in the expression to write this as:

$$\begin{aligned} J(\phi + v) &= \underbrace{\frac{1}{2} \int \nabla(\phi) \cdot \nabla(\phi) dx - \int f \cdot (\phi) dx}_{J(\phi)} \\ &+ \underbrace{\int \nabla(\phi) \cdot \nabla(v) dx - \int f \cdot (v) dx}_{\delta J_v(\phi)} \\ &+ \underbrace{\frac{1}{2} \int \nabla(v) \cdot \nabla(v) dx}_{\text{always positive}} \end{aligned}$$

Potential Flow – A Kinetic Energy Minimization Problem

- As a result, to minimize $J(\phi + v)$ (think flow kinetic energy), the solution ϕ has a first variation, $\delta J_v(\phi)$, equal to zero:

$$\underbrace{\int \nabla(\phi) \cdot \nabla(v) dx - \int f \cdot (v) dx}_{\delta J_v(\phi)} = 0 \quad (7)$$

- Incase it wasn't obvious, this is the **weak form** with the solution to the PDE, ϕ and the weighting function v .

Potential Flow – A Kinetic Energy Minimization Problem

- **Important result:** By solving the weak form of the equation, we are minimizing the potential energy of the flow (in this case).
- We saw that using FEM the weak form can be written as:

$$\frac{dE}{du} \simeq [A][\hat{u}] - [f] = [0] \quad (8)$$

- We can say the solution $[\hat{u}]$ minimizes the quadratic matrix equation (form) for the K.E.:

$$E = [u]^T [A][u] - [u]^T [f] \quad (9)$$

- Think of this as finding the minimum of an infinite dimensional paraboloid. The numerical solution is the “best” approximate solution that can be found given the basis functions/discretization (solution space) for the minimization of the Kinetic Energy.

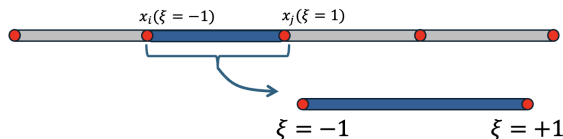
Questions?

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Example: Isoparametric Elements

- It is convenient to define a standard reference element (think: element computational domain) in parametric space.
 - For Isoparametric elements, **the same basis function interpolates the solution and the geometry mapping.**
 - We need to define a mapping between the *element* and *the physical*.
- For example, in 1D:



- Define following relationship between the reference element and the physical domain:

$$x(\xi = -1) = x_i$$

$$x(\xi = 1) = x_j$$

Example: Isoparametric Elements

- For an isoparametric element, the x value on the element can be written using the **same basis function interpolation** as is used for the solution:

$$\begin{aligned}x(\xi) &= \sum_{n=Nodes} x_n * N_n \\&= x_i * N_i + x_j * N_j \\&= x_i * \frac{1}{2}(1 - \xi) + x_j * \frac{1}{2}(1 + \xi)\end{aligned}$$

- Recall, the weak form requires terms such as $\frac{\partial u}{\partial x}$:

$$A_{i,j} \cdot u_j = \int_{x_i}^{x_j} \frac{\partial u_j}{\partial x} \cdot \frac{\partial v_i}{\partial x} dx \quad (10)$$

- We can write these using the chain rule as:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} \cdot \frac{\partial \xi}{\partial x} \quad (11)$$

Example: Isoparametric Elements

- From the previous slide:

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial \xi} \cdot \frac{\partial \xi}{\partial x} \quad (12)$$

- We are able to easily calculate the derivative $\frac{\partial u}{\partial \xi}$:

$$\begin{aligned} \frac{\partial u}{\partial \xi} \Big|_{Node i} &= \frac{-u_i}{2} \\ \frac{\partial u}{\partial \xi} \Big|_{Node j} &= \frac{u_j}{2} \end{aligned}$$

Example: Isoparametric Elements

- Now, let's look at the term $\frac{\partial \xi}{\partial x}$. Similar to Finite Difference Mappings:

$$\frac{\partial x}{\partial \xi} = \frac{(\partial x_i * \frac{1}{2}(1 - \xi) + x_j * \frac{1}{2}(1 + \xi))}{\partial x} = \frac{-x_i}{2} + \frac{x_j}{2} = \frac{h}{2} \quad (13)$$

- We can take the reciprocal to express:

$$\frac{\partial \xi}{\partial x} = \frac{2}{h} \quad (14)$$

- Combining these various calculations together:

$$\begin{aligned} \frac{\partial u}{\partial x} \Big|_{Node i} &= \underbrace{-u_i \frac{1}{2}}_{\frac{\partial u_i}{\partial \xi}} * \underbrace{\frac{2}{h}}_{\frac{\partial \xi}{\partial x}} = -u_i \frac{1}{h} \\ \frac{\partial u}{\partial x} \Big|_{Node j} &= \underbrace{u_j \frac{1}{2}}_{\frac{\partial u_j}{\partial \xi}} * \underbrace{\frac{2}{h}}_{\frac{\partial \xi}{\partial x}} = u_j \frac{1}{h} \end{aligned}$$

Example: Isoparametric Elements

- For the bilinear term, $a(u, v)$, which generates A-matrix entries:

$$\begin{aligned} A_e u_e &= \int_{x_i}^{x_j} \frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial x} dx \rightarrow \int_{\xi=-1}^{\xi=1} \left(\frac{\partial u_j}{\partial x} \cdot \frac{\partial v_i}{\partial x} \right) \frac{dx}{d\xi} \cdot d\xi \\ &\rightarrow \sum u_j \int_{\xi=-1}^{\xi=1} \left([-1/h, 1/h]^T \cdot [-1/h, 1/h] \right) \cdot \frac{h}{2} \cdot d\xi \end{aligned}$$

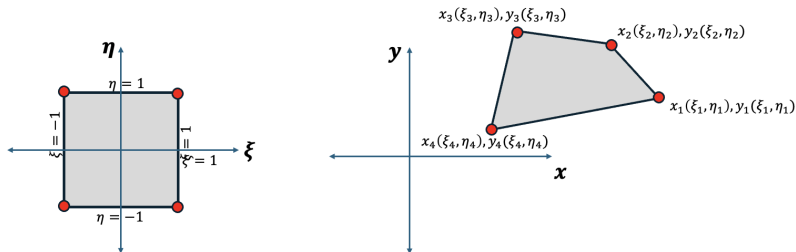
- Which results in the **same** element-matrix for the element with node i and node j:

$$A_e u_e = \begin{pmatrix} \frac{-1}{h_k} & \frac{1}{h_k} \\ \frac{1}{h_k} & \frac{-1}{h_k} \end{pmatrix} \begin{pmatrix} u_i \\ u_j \end{pmatrix} \quad (15)$$

- Which can be stamped into the matrix.
- Note:** Only the mapping term $(\frac{\partial x}{\partial \xi})$ needs to be calculated/determined for each (new) element that is considered.

Example: Isoparametric Elements

- For example, in 2D:



- The mapping will be similar to the 2-D Finite Difference Mappings. A similar process is used to that demo'd in 1D.

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Let's talk basis/shape functions

- Basis/shape functions are used to represent the solution.
- So far, we have used linear basis functions in 1D and 2D (could generalize to 3D).
- What were some of the criteria (T.J.R. Hughes):
 - 1 Basis/shape functions must be smooth inside each element Ω_e
 - At least C^1 for our Poisson problems
 - Means the basis function must have continuous derivative values on each element.
 - 2 Continuous function value across the element boundaries (Γ_e).
 - We permit jumps in the derivatives at the boundaries, but the value of the solution must be continuous from one element to the next.
 - 3 Complete
 - Requires that the *"element interpolation function is capable of exactly representing an arbitrary linear polynomial when the nodal degrees of freedom are assigned values in accordance with it."* (Page 111 of T.R.J. Hughes).
 - This is because as we refine the elements used to represent the solution, the solution will tend towards a combination of constant + linear monomial.

Example: Some 1D Shape Functions

- In one dimension, we already saw linear basis functions.
- Parametrized into element coordinates:

$$N_{\textcolor{red}{1}(\xi)} = \frac{\xi - \xi_2}{\textcolor{red}{\xi_1} - \xi_2} \rightarrow \xi_1 = -1 \text{ and } \xi_2 = +1 \rightarrow \textcolor{red}{\frac{1}{2}}(\textcolor{red}{1} - \xi)$$

$$N_{\textcolor{blue}{2}(\xi)} = \frac{\xi - \xi_1}{\textcolor{blue}{\xi_2} - \xi_1} \rightarrow \xi_1 = -1 \text{ and } \xi_2 = +1 \rightarrow \textcolor{blue}{\frac{1}{2}}(\textcolor{blue}{1} + \xi)$$

- We can see that these are first order Lagrange Polynomials.

Example: Some 1D Shape Functions

- The expression for Lagrange Polynomials (L_a^p) of order p on **node a** is (i.e., the shape function (N_a)):

$$L_a^p(\xi) = N_a(\xi) = \frac{\prod(\xi - \xi_b)}{\prod(\xi_a - \xi_b)}, \text{ for products over } b = 1, b \neq a \text{ to } n$$

- So, the second order (quadratic) Lagrange Polynomial 1D Nodal Basis Functions would be:

$$N_{1(\xi)} = \frac{(\xi - \xi_2)(\xi - \xi_3)}{(\xi_1 - \xi_2)(\xi_1 - \xi_3)} \rightarrow \xi_1 = -1 \text{ and } \xi_2 = +1 \rightarrow \frac{\xi}{2}(\xi - 1)$$

$$N_{2(\xi)} = \frac{(\xi - \xi_1)(\xi - \xi_3)}{(\xi_2 - \xi_1)(\xi_2 - \xi_3)} \rightarrow \xi_1 = -1 \text{ and } \xi_2 = +1 \rightarrow (1 - \xi^2)$$

$$N_{3(\xi)} = \frac{(\xi - \xi_1)(\xi - \xi_2)}{(\xi_3 - \xi_1)(\xi_3 - \xi_2)} \rightarrow \xi_1 = -1 \text{ and } \xi_2 = +1 \rightarrow \frac{\xi}{2}(\xi + 1)$$

Example: Some 1D Shape Functions

- Let's try to construct these 1D quadratic basis functions in MATLAB.

$$N_1(\xi) = \frac{\xi}{2}(\xi - 1)$$

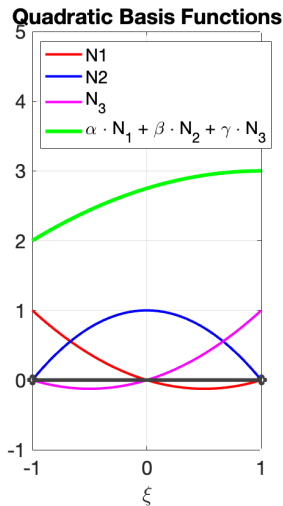
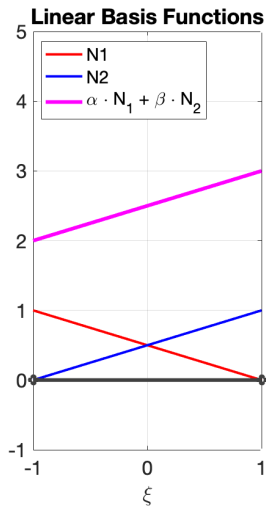
$$N_2(\xi) = (1 - \xi^2)$$

$$N_3(\xi) = \frac{\xi}{2}(\xi + 1)$$

- Also, determine the expressions for the first derivatives ($\frac{\partial N_1}{\partial \xi}$, and $\frac{\partial N_2}{\partial \xi}$).

Example: Some 1D Shape Functions

- Both linear and quadratic 1-D basis functions are shown.



Example: Some 2D Shape Functions

- We can extend this idea to two dimensions by writing the shape functions for a bilinear quadrilateral element as the product of first order Lagrange Polynomials:

$$N_a(\xi, \eta) = L_b^1(\xi) * L_c^1(\eta) \quad (16)$$

- With:

$$\begin{aligned} N_1(\xi) &= \frac{\xi - \xi_2}{\xi_1 - \xi_2} \rightarrow \xi_1 = -1 \text{ and } \xi_2 = +1 \rightarrow \frac{1}{2}(1 - \xi) \\ N_2(\xi) &= \frac{\xi - \xi_1}{\xi_1 - \xi_2} \rightarrow \xi_1 = -1 \text{ and } \xi_2 = +1 \rightarrow \frac{1}{2}(1 + \xi) \end{aligned}$$

Example: Some 2D Shape Functions

- This results in the following 4 basis functions on a 2D quadrilateral:

$$N_1(\xi, \eta) = \frac{1}{2}(1 - \xi) * \frac{1}{2}(1 - \eta) = \frac{1}{4}(1 - \xi) * (1 - \eta)$$

$$N_2(\xi, \eta) = \frac{1}{2}(1 + \xi) * \frac{1}{2}(1 - \eta) = \frac{1}{4}(1 + \xi) * (1 - \eta)$$

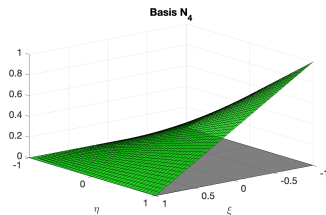
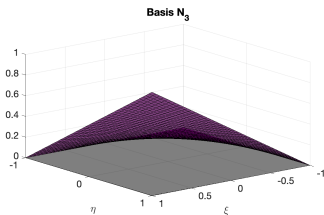
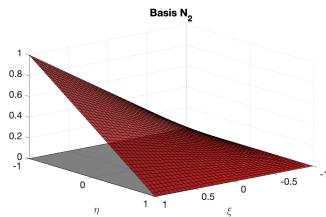
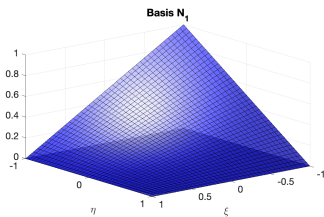
$$N_3(\xi, \eta) = \frac{1}{2}(1 + \xi) * \frac{1}{2}(1 + \eta) = \frac{1}{4}(1 + \xi) * (1 + \eta)$$

$$N_4(\xi, \eta) = \frac{1}{2}(1 - \xi) * \frac{1}{2}(1 + \eta) = \frac{1}{4}(1 - \xi) * (1 + \eta)$$

- Try this out in MATLAB (with plots).

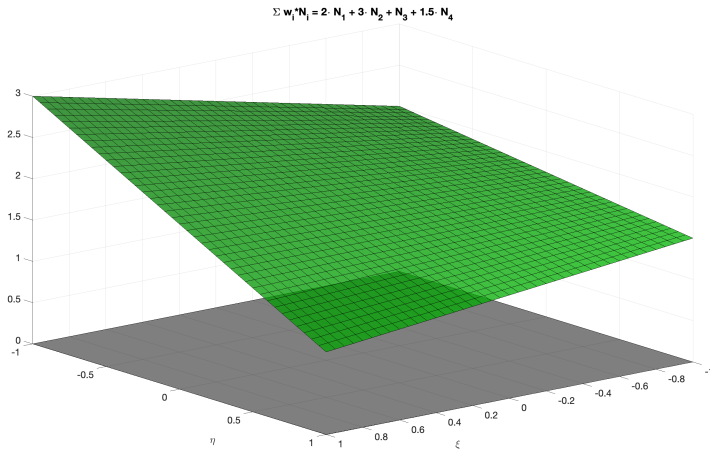
Example: Some 2D Shape Functions

- There are 4 bi-linear basis functions on a 2D quadrilateral element:



Example: Some 2D Shape Functions

- Weighted and added together to represent a bi-linear function on the element:



Example: Some 2D Shape Functions

- Similarly, we can define a 9-noded quadrilateral element with 2nd order Lagrange Polynomials as basis functions:

$$N_a(\xi, \eta) = L_b^2(\xi) * L_c^2(\eta) \quad (17)$$

- With:

$$N_1(\xi) = \frac{(\xi - \xi_2)(\xi - \xi_3)}{(\xi_1 - \xi_2)(\xi_1 - \xi_3)} \rightarrow \xi_1 = -1 \text{ and } \xi_2 = +1 \rightarrow \frac{\xi}{2}(\xi - 1)$$

$$N_2(\xi) = \frac{(\xi - \xi_1)(\xi - \xi_3)}{(\xi_2 - \xi_1)(\xi_2 - \xi_3)} \rightarrow \xi_1 = -1 \text{ and } \xi_2 = +1 \rightarrow (1 - \xi^2)$$

$$N_3(\xi) = \frac{(\xi - \xi_1)(\xi - \xi_2)}{(\xi_3 - \xi_1)(\xi_3 - \xi_2)} \rightarrow \xi_1 = -1 \text{ and } \xi_2 = +1 \rightarrow \frac{\xi}{2}(\xi + 1)$$

- Let's try this out now!

Example: Some 2D Shape Functions

- This results in 9 × quadratic basis functions on a 2D quadrilateral (here are some examples):

$$N_{corner1}(\xi, \eta) = \frac{\xi * \eta}{4} (\xi - 1) * (\eta - 1)$$

$$N_{middle5}(\xi, \eta) = \frac{\eta}{2} (1 - \xi^2) * (\eta - 1)$$

$$N_{midside9}(\xi, \eta) = (1 - \xi^2) * (1 - \eta^2)$$

- Try this out in MATLAB (with plots).

Example: 2D Shape Functions: 9-Node Quadratic on a 4 Sided Element

- Here are the 3 main basis functions on a 9 node element (corner, mid-edge, middle):

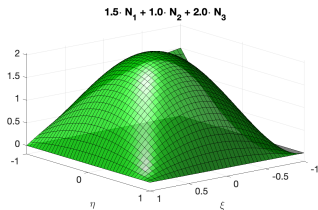
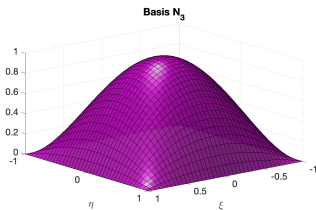
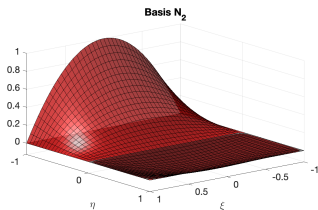
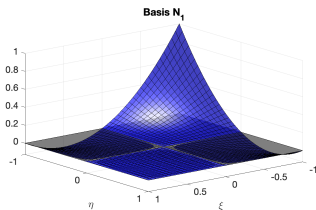


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Gauss Quadrature for Integration

- The **weak form** requires integration over element.
- As the functions being integrated increase in complexity, numerical integration is often used.
 - This is especially true for the RHS or source term integrals.
- Gauss Quadrature methods are often used for integration:

$$\int \int_{\Omega^e} f(x, y) dA_e = \sum_{n=1}^{NP} w_i * f(x_i, y_i) \quad (18)$$

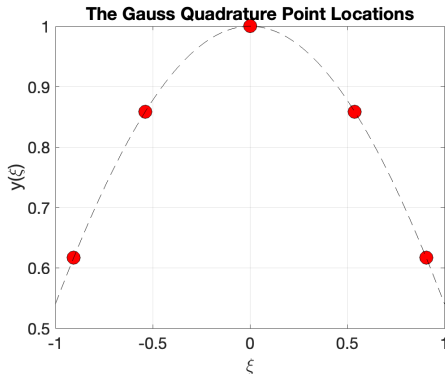
- Which requires the evaluation of the integrand $f(x, y)$ at the Gauss Quadrature points.
- These points are then weighted and summed together.
- Let's try an example integral using Gauss Quadrature.

Gauss Quadrature for Integration

- Consider 5-point Gauss Quadrature for evaluating the integral:

$$\int_{\xi=-1}^{\xi=+1} \cos(x) dx \approx \sum_{n=1}^5 w_n * f(x_n) \quad (19)$$

- The plot of the points used in this integration is shown below.



Gauss Quadrature for Integration

- Consider the following integral:

$$\int_{\xi=-1}^{\xi=1} \int_{\eta=-1}^{\eta=1} \sin(2\pi\xi) * \sin(2\pi\eta) * \frac{\xi * \eta}{4} (\xi - 1) * (\eta - 1) d\eta d\xi \quad (20)$$

- Look on wikipedia for a 5-point Gauss Quadrature scheme to use:

https://en.wikipedia.org/wiki/Gauss-Legendre_quadrature

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Setting up and Coding Transient 1D Heat Conduction Problem

- In Finite Differences we solved the following:

$$\frac{\partial T}{\partial t} + \kappa \frac{\partial^2 T}{\partial x^2} = f(x) = 0 \quad (21)$$

- With initial conditions in the domain:

$$T(0) = 0 \quad (22)$$

- And boundary conditions:

$$T(x=0) = 0$$

$$T(x=1) = 1$$

Setting up and Coding Transient 1D Heat Conduction Problem

- Let's multiply the strong form by a test function v_i and integrate over the domain:

$$\int_{\Omega} v_i \frac{\partial T}{\partial t} dx + \kappa \int_{\Omega} v_i \frac{\partial^2 T}{\partial x^2} dx = \int_{\Omega} v_i f(x) = 0 \quad (23)$$

- Now let's integrate the spatial term by parts:

$$\int_{\Omega} v_i \frac{\partial T}{\partial t} dx + \left[v_i \cdot \frac{\partial T}{\partial n} \right]_{\Gamma} - \kappa \int_{\Omega} \frac{\partial v_i}{\partial x} \frac{\partial T}{\partial x} dx = 0 \quad (24)$$

- Introduce basis function representations:

$$T(x) = \sum T_j \phi_j \quad \text{and} \quad v_i = \phi_i \quad (25)$$

Setting up and Coding Transient 1D Heat Conduction Problem

- Let's multiply the strong form by a test function v_i and integrate over the domain:

$$\int_{\Omega} v_i \frac{\partial T}{\partial t} dx + \kappa \int_{\Omega} v_i \frac{\partial^2 T}{\partial x^2} dx = \int_{\Omega} v_i f(x) = 0 \quad (26)$$

- Now let's integrate the spatial term by parts:

$$\sum \frac{\partial T_j}{\partial t} \int_{\Omega} \phi_i \cdot \phi_j \cdot dx + \left[\phi_i \cdot \frac{\partial T}{\partial n} \right]_{\Gamma} - \sum T_j \kappa \int_{\Omega} \frac{\partial \phi_i}{\partial x} \frac{\partial \phi_j}{\partial x} dx = 0 \quad (27)$$

Setting up and Coding Transient 1D Heat Conduction Problem

- We can see the **first** integral will produce an element-based matrix – the **capacity** matrix:

$$[C] = \int_{\Omega} \phi_i \cdot \phi_j \cdot dx = \begin{pmatrix} \frac{h}{3} & \frac{h}{6} \\ \frac{h}{6} & \frac{h}{3} \end{pmatrix} \quad (28)$$

- Similarly, the second integral produces an element based **stiffness matrix**:

$$[K] = \int_{\Omega} \frac{\partial \phi_i}{\partial x} \cdot \frac{\partial \phi_j}{\partial x} \cdot dx = \begin{pmatrix} \frac{1}{h} & -\frac{1}{h} \\ -\frac{1}{h} & \frac{1}{h} \end{pmatrix} \quad (29)$$

Setting up and Coding Transient 1D Heat Conduction Problem

- This takes the matrix form for an arbitrary element:

$$\begin{pmatrix} \frac{h}{3} & \frac{h}{6} \\ \frac{h}{6} & \frac{h}{3} \end{pmatrix} \begin{pmatrix} \frac{dT_i}{dt} \\ \frac{dT_j}{dt} \end{pmatrix} + \begin{pmatrix} \frac{1}{h} & -\frac{1}{h} \\ -\frac{1}{h} & \frac{1}{h} \end{pmatrix} \begin{pmatrix} T_i \\ T_j \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (30)$$

- Once assembled into the global stiffness and capacity matrices, the FEM discretization is:

$$[C][\dot{T}] + [K][T] = 0 \quad (31)$$

- Now the task becomes discretizing $\dot{T}$ which can be accomplished as an ODE solve (e.g.: forward or backward Euler among many methods):

$$[C] \left(\frac{T_{t+\Delta t} - T_t}{\Delta T} \right) + [K][T] = 0 \quad (32)$$

Setting up and Coding Transient 1D Heat Conduction Problem

- Try programming this now!
- Try implicit and explicit
- Careful with boundary conditions

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FEM: Where to Learn More

- MECH.5130 Theory of Finite Element Analysis, UML course.
- T.R.J. Hughes,
The Finite Element Method: Linear Static and Dynamic Finite Element
cheaper, solid start.
- K.J. Bathe, Finite Element Procedures, Comprehensive
solids/structural FEM
- MIT Open Courseware: 2.002
- G. Strang, An Analysis of the Finite Element Method – Older /classic
book, math theory.